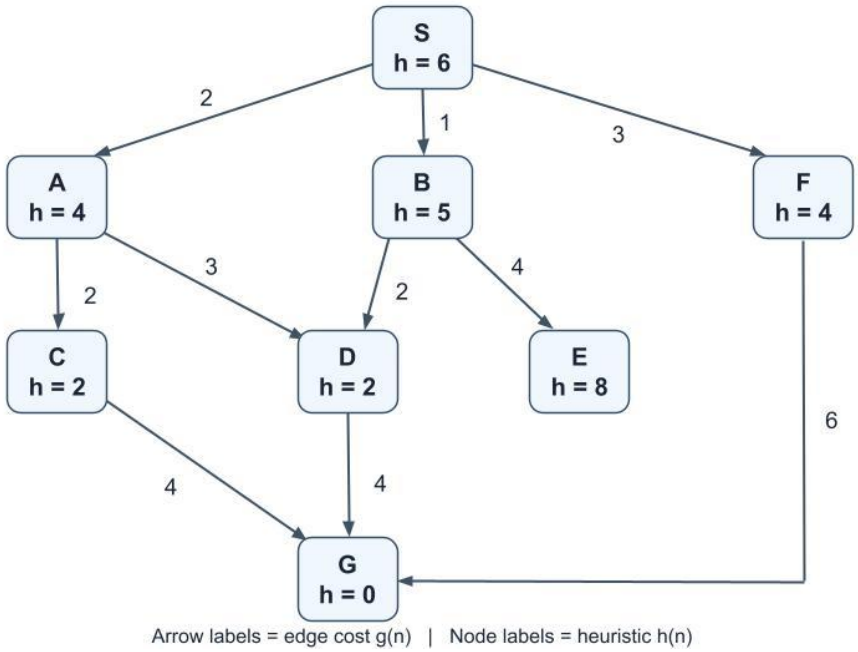
# SC5002 Lab Assignment 1 Report

## Part 1: Classical Search & Problem Solving

### Task 1.1: Graph Traversal & Heuristic Search




1. **Path Search**

| Search Algorithm | Complete Path | Total Cost |
|---|---|---|
| Breadth-First Search (BFS) | S → F → G | 3+6=9 |
| A* Search | S → B → D → G | 1+2+4=7 |

2. **Evaluation table (A*Search):**

| Steps | OPEN | | | | CLOSED | |
|---|---|---|---|---|---|---|
| | Node | g(n) | h(n) | f(n) | Node | Parent |
| Initial | S | 0 | 6 | 6 | | |
| Iteration 1 (S) | A | 2 | 4 | 6 | S | — |
| | B | 1 | 5 | 6 | | |
| | F | 3 | 4 | 7 | | |
| Iteration 2 (A) | B | 1 | 5 | 6 | S | — |
| | C | 4 | 2 | 6 | A | S |
| | D | 5 | 2 | 7 | | |
| | F | 3 | 4 | 7 | | |
| Iteration 3 (B) | D | 3 | 2 | 5 | S | — |

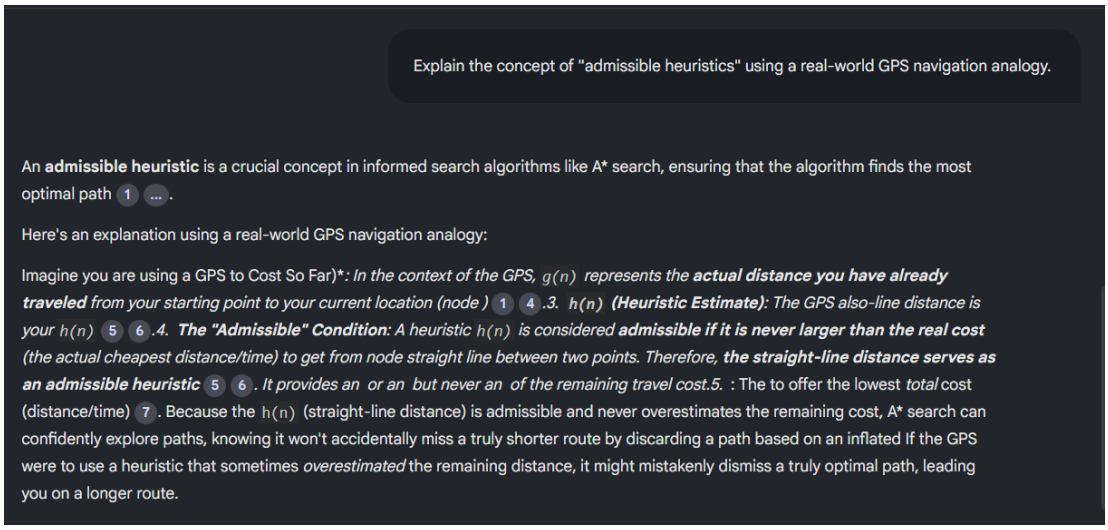
| | | | | | | |
|---|---|---|---|---|---|---|
| | C | 4 | 2 | 6 | A | S |
| | F | 3 | 4 | 7 | B | S |
| | E | 5 | 8 | 13 | | |
| Iteration 4 (D) | C | 4 | 2 | 6 | S | — |
| | F | 3 | 4 | 7 | A | S |
| | G | 7 | 0 | 7 | B | S |
| | E | 5 | 8 | 13 | D | B |
| Iteration 5 (C) | F | 3 | 4 | 7 | S | — |
| | G | 7 | 0 | 7 | A | S |
| | E | 5 | 8 | 13 | B | S |
| | | | | | D | B |
| | | | | | C | A |
| Iteration 6 (F) | G | 7 | 0 | 7 | S | — |
| | E | 5 | 8 | 13 | A | S |
| | | | | | B | S |
| | | | | | D | B |
| | | | | | C | A |
| | | | | | F | S |
| Iteration 7 (G) | E | 5 | 8 | 13 | S | — |
| | | | | | A | S |
| | | | | | B | S |
| | | | | | D | B |
| | | | | | C | A |
| | | | | | F | S |
| | | | | | G | D |

A* Result:
S → B → D → G Total cost 1 + 2 + 4 = 7.

Remark:
When nodes had equal f(n) values, I chose them alphabetically, which explains selecting A before B and F before G.

## LEARN WITH AI TASK 1.1

### AI Prompting Log (Gemini Notebook)



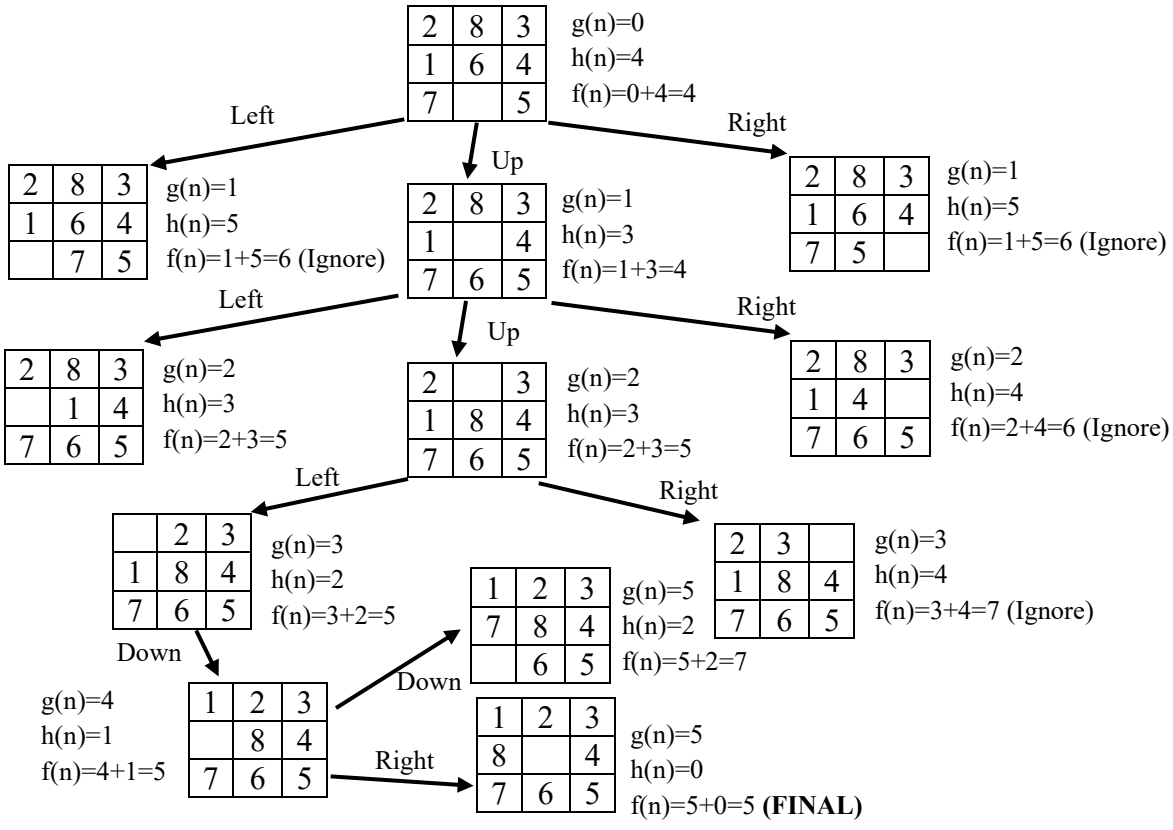

In A* search, g(n) is the actual cost from the starting point to the current node, while h(n) estimates the remaining cost to the goal. A heuristic is admissible when it never overestimates the cheapest remaining cost. For example, a GPS can use straight-line distance when finding the shortest route because travelling along roads cannot give a shorter distance than the straight line. For the question, h(B)=5 is admissible. The cheapest route from B to G is:

$$B \rightarrow D \rightarrow G \quad \text{Cost: } 2 + 4 = 6$$

Since $h(B)=5 \leq 6$, it underestimates the true cost and therefore it is admissible.

## Task 1.2: 8-Puzzle Problem Solving

Assume g(n)= depth of node and h(n)=number of misplaced tiles:



**LEARN WITH AI TASK 1.2**

**AI Prompt (Gemini Notebook):**

I am solving an 8-puzzle problem using A* search, where g(n) is the depth of the node and h(n) is the number of misplaced tiles. I have calculated the values for each possible move. I have attached my working in the source. Check my calculations and node-selection decisions. Identify any errors or steps that need reconsideration, but do not solve the problem for me.

From the feedback that Gemini gave, Gemini stated that my g(n) values were correct and most f(n) calculations were consistent with the values provided. It questioned several heuristic counts and recommended reviewing the node selections, but its heuristic feedback incorrectly included the blank tile. It also identified the final arithmetic error. My working showed g(n)=5, h(n)=0, and incorrectly calculated f(n)=5+0=1. That was my careless calculation error. I accepted Gemini's suggestion and corrected it to f(n)=5. However, I rejected its suggestion to increase the initial heuristic from 4 to 5 because the misplaced-tile heuristic excludes the blank, leaving only tiles 1, 2, 6 and 8 misplaced.

# Part 2: Applied Machine Learning with AI Copilot

**Task 2.1: Dataset Setup & AI Feature Analysis**

**AI Prompt (Gemini Notebook):**

```
I am a non-programmer studying Machine Learning. Here are the first 5 rows from my dataset:
=== TASK 2.1: Loading California Housing Dataset ===
First 5 rows of dataset (Copy these for AI Prompt 2.1):
MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude

20046  1.6812    25.0  4.192201   1.022284    1392.0  3.877437    36.06
3024   2.5313    30.0  5.039384   1.193493    1565.0  2.679795    35.14
15663  3.4801    52.0  3.977155   1.185877    1310.0  1.360332    37.80
20484  5.7376    17.0  6.163636   1.020202    1705.0  3.444444    34.28
9814   3.7250    34.0  5.492991   1.028037    1063.0  2.483645    36.62


   Longitude  MedHouseVal


20046    -119.01    0.47700
3024     -119.46    0.45800
15663    -122.44    5.00001
20484    -118.72    2.18600
9814     -121.93    2.78000
Identify: (1) The continuous target variable, (2) The input features, and (3) Why
numerical feature scaling (StandardScaler) helps linear regression."
```

**Summary of Gemini's response:**

- **Continuous target variable:** MedHouseVal is the median house value that the model predicts. Since it is a continuous numerical value, this is a regression problem.
- **Input features:** MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude and Longitude.
- **Feature scaling:** These features have different units and ranges. StandardScaler subtracts each feature's training mean and divides by its training standard deviation, putting the features on comparable scales.

- **Benefits:** Scaling can improve numerical conditioning and makes coefficients easier to compare. It is especially useful for gradient-based learning and regularised models such as Ridge and Lasso.
- **Clarification:** Scikit-learn's standard LinearRegression uses a least-squares solver rather than gradient descent. Scaling does not automatically improve its prediction accuracy, and larger feature values do not necessarily produce larger coefficients.

**Task 2.2: Model Execution & Alpha Tuning**

**AI Prompt (Gemini Notebook):**

My 5-fold cross-validation $R^2$ score for Linear Regression is 0.6457, but Ridge Regression with alpha=10 scored 0.6443. Explain to a non-technical manager why Ridge performed differently and why cross validation gives a more reliable metric than a single train/test split.

**Evaluation Scores Table:**

| Model | alpha (α) | Mean 5-Fold $R^2$ Score |
|---|---|---|
| Linear Regression | - | 0.6457 |
| Ridge Regression | 0.1 | 0.6457 |
| Ridge Regression | 10 | 0.6443 |
| Ridge Regression | 1000 | 0.3574 |

**Summarised AI explanation:**

**Why Ridge performed differently?**

- **Linear Regression** fits the housing data by minimising the squared differences (errors) between predicted and actual house values.
- **Ridge Regression** also fits the data but adds a penalty for large coefficients. This encourages more stable predictions and can reduce overfitting. Increasing alpha strengthens this penalty, although stronger regularisation does not always improve performance.
- **Comparison of the reported results:** Linear Regression scored 0.6457, while Ridge with α = 10 scored 0.6443. Linear Regression therefore performed slightly better in this evaluation, with a difference of only **0.0014**. Without the individual fold scores, this small difference is insufficient to establish a meaningful performance advantage.

**Why cross validation gives a more reliable metric than a single train/test split?**

- **How 5-fold cross-validation works:** The data is divided into five folds. Each model is trained on four folds and evaluated on the remaining fold. This is repeated five times, allowing each fold to serve as validation data once, and the five $R^2$ scores are averaged.
- **Why it is more reliable:** Evaluating several held-out subsets makes the result less dependent on one unusually easy or difficult train/test split. It provides a broader assessment of how the model performs on unseen examples.
- **Alpha tuning:** The scores for α = 0.1, 10 and 1000 should be compared to identify which tested penalty gives the highest mean validation $R^2$.

**Task 2.3: Visual Analysis & Overfitting Insights**

**AI Prompt (Gemini Notebook):**

Why does setting the Ridge penalty alpha extremely high (e.g., 1000) cause model performance ($R^2$ score) to drop significantly?"

**Summarised AI explanation:**

The AI's explanation helped me understand why the Ridge penalty needs to be chosen carefully. In my results, the mean cross-validation $R^2$ decreased from 0.6457 at alpha = 0.1 to 0.6443 at alpha = 10, then dropped considerably to 0.3574 at alpha = 1000. This large drop is consistent with underfitting, where an excessive penalty restricts the model's ability to capture useful relationships. A weak penalty however may leave the model more likely to learn noise and overfit. I now understand that choosing alpha involves balancing these two risks, using cross-validation to identify which tested setting performs best on unseen data.

GITHUB Repository:
https://github.com/Ambrose3112/SC5002-Lab-Assignment-1